

Recipe Traffic Prediction Project

Predicting High-Traffic Recipes for Business Optimization

Eleanor Koh | 26 Jan 2025



Project Overview and Business Goals

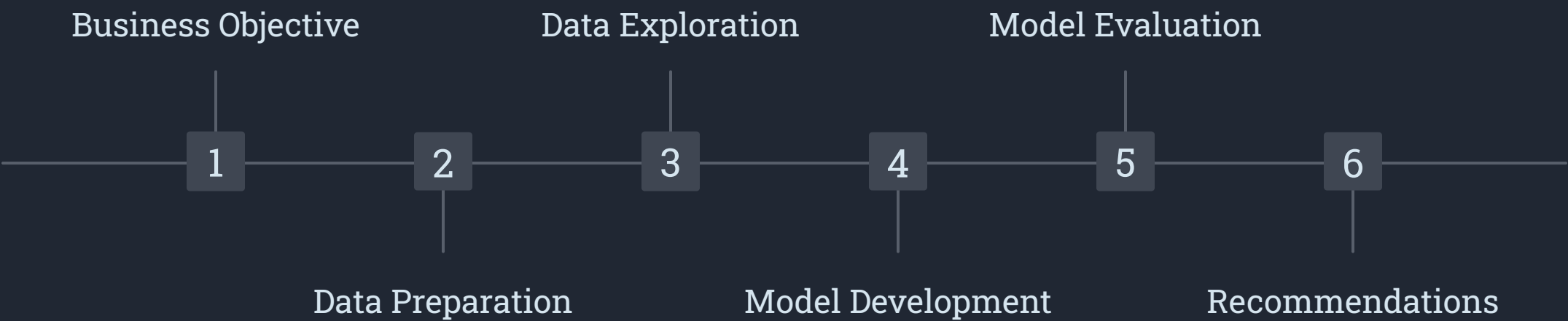
Business Goal

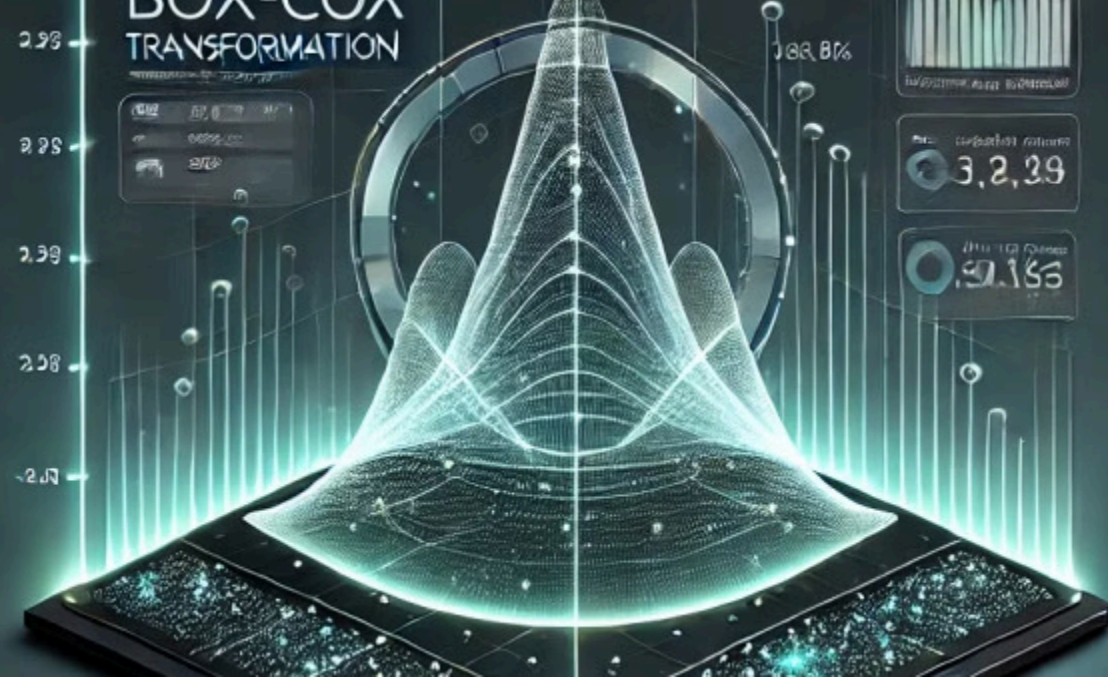
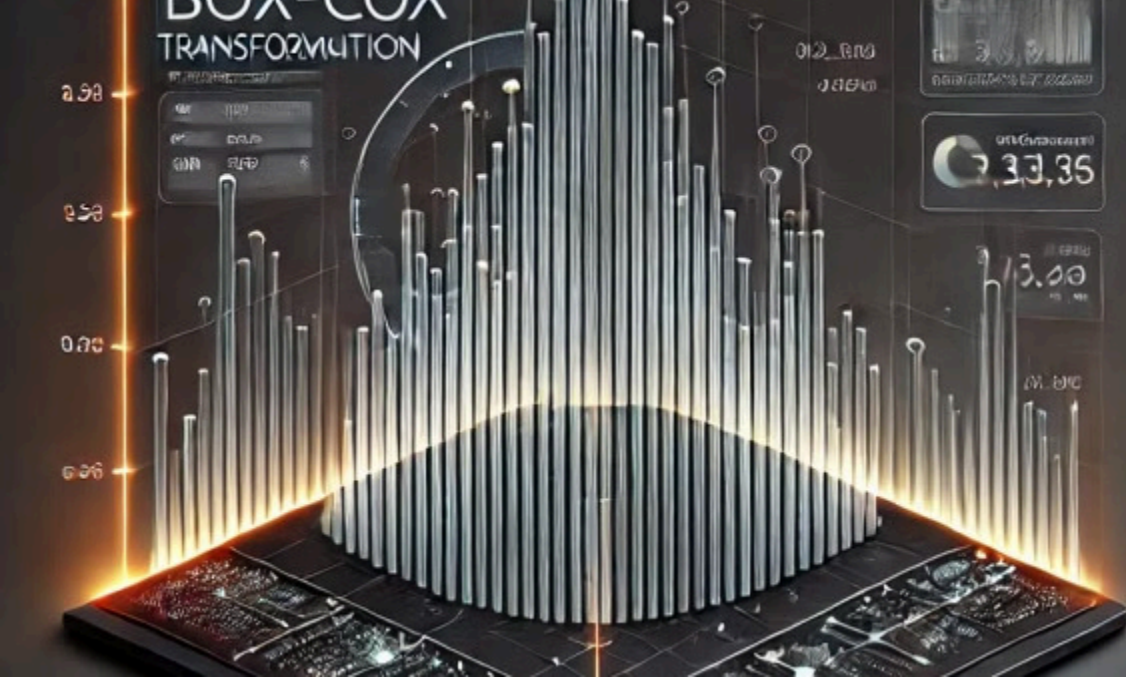
Increase user engagement with high-traffic recipes.

Objective

Predict popular recipes with at least 80% precision, minimizing unpopular recipes.

Project Overview





Data Preparation

Missing Values

52 missing entries (~5.5%) in nutrient columns imputed using the median.

Outlier Analysis

Outliers retained, aligned with domain knowledge.

Data Transformation

Box-Cox transformation applied to address skewed distributions.

Exploratory Data Analysis

Key Findings

1. Significant association between recipe categories vs traffic.
2. Strong correlation with traffic:
 - Pork, Potato, Vegetable
3. Weak but significant correlation with traffic:
 - Calories, Protein, Sugar

Insights and Actions

Strong categorical correlation as strong predictors of high traffic.

Engineered "*Nutrients per serving*" column

Model Development



Logistic Regression

A simple and interpretable linear model used for binary classification.



Random Forest

An ensemble method combining multiple decision trees for improved accuracy.



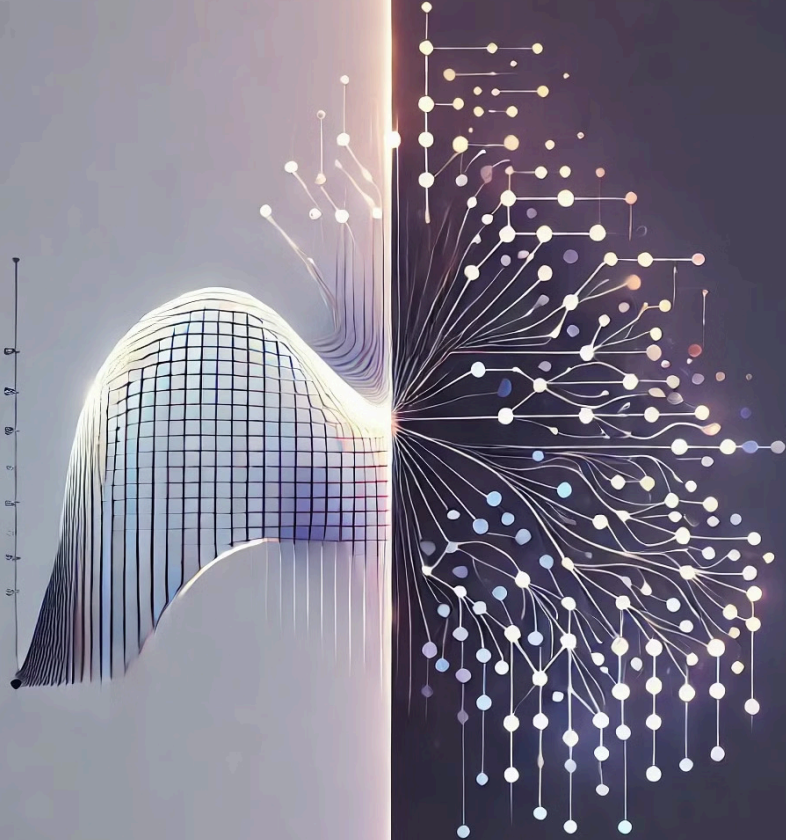
Support Vector Machine

A powerful model that finds optimal hyperplanes to separate data points.



Gradient Boosting

An ensemble method that builds trees sequentially, correcting errors of previous trees.



Model Evaluation

Metric	Logistic Regression (Original Dataset)	Gradient Boosting (Nutrients by Servings)
Precision	87%	74%
F1-Score	0.815	0.828

Logistic Regression:

- **Precision:** Aligns with the business goal of **minimizing false positives**.
- **Interpretability:** Clear insights into **feature importance**, aiding decision-making

Gradient Boosting (Tuned):

- **F1-Score:** **0.828** (Highest overall performance)
- **Recall:** Excels at identifying **true positives**, minimizing **false negatives**.

Feature Importance

Top Predictive Features (Original Dataset):

- Vegetable (importance score: 2.32)
- Potato (importance score: 1.72)
- Pork (importance score: 1.30)

Key Insight:

These three categories demonstrated **statistically significant associations** with recipe traffic during Exploratory Data Analysis (EDA).



Recommendations

Model: Logistic Regression

- 1 Highest precision**
Ensures accurate identification of high-traffic recipes
- 2 Intuitive insights**
Supports strategic business decisions

Business Actions:

- 1 Optimize Recipes**
Promote high-traffic categories - **Vegetable, Potato, Pork.**
- 2 Data Enrichment**
Enrich data by including fat and collecting user preference trends.



Takeaways & Future Work

87%

Precision

Logistic Regression

3

Predictors

Vegetables, Potato, Pork

Fat

Data Enrichment